

Tutorial Sheet - Topic 03

Programming Fundamentals 2

Revision (ArrayLists)

Part A: Understanding ArrayLists

1. Declaring and Creating ArrayLists

Write Java code for each of the following:

1. Declare and create an `ArrayList<String>` called `names`.
 2. Declare and create an `ArrayList<Integer>` called `scores`.
 3. Add three names to the `names` `ArrayList`.
 4. Add three integer values to the `scores` `ArrayList`.
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2. Predicting Output

Without running the code, write down exactly what is printed:

```
ArrayList<Integer> numbers = new ArrayList<Integer>();  
numbers.add(5);  
numbers.add(10);  
numbers.add(15);
```

```
System.out.println(numbers.get(1));  
System.out.println(numbers.size());
```

Answer the following:

- a. What is printed on the first line?
 - b. What is printed on the second line?
-

3. Understanding Indexing

Consider the following code:

```
ArrayList<String> colours = new ArrayList<String>();  
colours.add("Red");  
colours.add("Green");  
colours.add("Blue");
```

Answer the following:

- What element is stored at index 0?
 - What element is stored at index 2?
 - What happens if `colours.get(3)` is executed?
-

4. Correcting Logic

The following code is intended to print all elements in an ArrayList, but it contains an error.

```
ArrayList<String> animals = new ArrayList<String>();  
animals.add("Dog");  
animals.add("Cat");  
animals.add("Horse");  
  
for (int i = 0; i <= animals.size(); i++) {  
    System.out.println(animals.get(i));  
}
```

- Explain what is wrong with the loop condition.
 - Correct the code.
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Part B: Using Loops with ArrayLists

5. Looping with a for Loop

Write Java code to:

1. Print all elements in the names ArrayList using a for loop.
 2. Print all values in the scores ArrayList using a for loop.
-

6. Looping with a for-each Loop

Given the following ArrayList:

```
ArrayList<String> modules = new ArrayList<String>();  
modules.add("Programming");  
modules.add("Databases");  
modules.add("Web Development");
```

1. Write a for-each loop to print all module names.
 2. Explain one limitation of using a for-each loop with an ArrayList.
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7. Removing Elements

Consider the following code:

```
ArrayList<Integer> values = new ArrayList<Integer>();  
values.add(10);  
values.add(20);  
values.add(30);  
values.remove(1);
```

Answer the following:

- a. Which value is removed from the list?
 - b. What value is now stored at index 1?
 - c. What is the size of the ArrayList after removal?
-

Part C: Applying ArrayLists

8. Checking for Empty Lists

Write Java code to check if an `ArrayList<Product>` called `products` is empty. Use the most appropriate method.

9. Finding the Lowest Value

You are given the following `ArrayList` containing **five integer values**:

prices

25	40	15	60	30
0	1	2	3	4

(Assume the values are stored in an `ArrayList<Integer>` in the order shown.)

Part A – Pseudocode

Write pseudocode that describes the steps needed to find the **lowest value** in the `ArrayList`.

Your pseudocode should: - Make an initial assumption - Compare values one at a time - Update the lowest value when needed

Part B – Java Code

Using your pseudocode as a guide, write Java code to find and print the lowest value in the `prices ArrayList`.

10. Thinking Challenge

Answer the following:

1. Give one advantage of using an ArrayList instead of an array in a program such as a Store class.
2. Why is it no longer necessary to keep a separate total variable when using an ArrayList?